ENGINEERING DESIGN EVENT

The objective of the Engineering Design Event is to evaluate the engineering effort that has gone into the vehicle's design, how effectively the design meets market requirements, and how well it simulates real-world engineering projects and their associated challenges. The team that demonstrates the best application of engineering principles to achieve its design goals, along with the strongest understanding of its vehicle's design, will be declared the winner of the Engineering Design Event.

Note: Teams are reminded that BAJA SAEINDIA is an *engineering design competition.* In the Engineering Design Event, teams are evaluated based on their engineering design efforts. Components and systems incorporated as finished items are not judged as student-designed units; instead, they are assessed only on the team's rationale for their selection and application. For example, teams that design and fabricate their own shock absorbers are evaluated on both the design of the shock absorber and its integration within the suspension system. Teams using commercially available shock absorbers are evaluated solely on their selection and application within the suspension system.

Engineering Design Docket:

Engineering Design Docket consists of the following documents, which must be submitted by the teams to be eligible for design presentation. The template for all the design docket documents is uploaded on BAJA SAEINDIA Forum.

Design Report -

A maximum of 5 pages of text, 3 pages of drawings (Vehicle views- Front, Side & Top) & Annexure for Design calculations (Limited to 10 Pages) are allowed in the design report for **mBAJA Teams**.
A maximum of 6 pages of text, 3 pages of drawings (Vehicle views- Front, Side & Top) & Annexure for Design calculations (Limited to 10 Pages) are allowed in the design report for **eBAJA Teams**.

Design Specification Sheet –

A completed **BAJA SAEINDIA Design Specification Sheet** must be submitted by all teams. The template provided should not be altered or reformatted in any way before submission. The Design Judges understand that final design refinements and vehicle development may lead to slight variations between the submitted specifications and the completed vehicle. For parameters subject to tuning, teams may indicate an anticipated range of values as appropriate.

DFMEA and DVP&R -

The **Design Failure Mode and Effects Analysis (DFMEA)** and **Design Validation Plan & Report (DVP&R)** are essential components of the BAJA SAEINDIA Design Docket submission, as they contribute significantly to enhancing vehicle reliability, safety, and overall performance. In addition, this year, teams are required to include images of the validations performed at the vehicle, subsystem, or component level.

Design Improvement (For Returning Teams) -

Teams whose vehicles participated in BAJA SAEINDIA 2025 competition must document the improvements in their current design compared to the previous year, using the template to be published on the BAJA SAEINDIA Forum. As part of the Engineering Design Event, judges will evaluate this documentation, and if the design changes are **(A)** not significant, **(B)** lack detailed analysis, or **(C)** are insufficiently documented, a penalty will be applied to the design score.

Additionally, this year, teams are required to perform a **Root Cause Analysis (RCA)** using a *Fishbone Diagram* to identify **issues or failures** encountered in the previous year related to vehicle performance, components, or assembly. Teams are advised to conduct the RCA by comparing the previous year's specifications with the current year's, using the template provided in the last slide.

Engineering Design Docket Submission Guidelines:

Teams are required to follow the following procedure to submit the Design Docket to BAJA SAEINDIA.

Submission Link	**BAJA SAEINDIA Website**		
	To be done on the BAJA SAEINDIA Website in the submission section in the team profile.		
Last Date of Submission	**21st December 2025, 23:59 IST**		
Document	**Submission**	**File Type**	**File Name**
Design Report	Compulsory	.pdf	TeamID_TeamName_Design_Report.pdf
Design Specification Sheet	Compulsory	.xlsx	TeamID_TeamName_Spec_Sheet.xlsx
DVP&R and DFMEA	Compulsory	.xlsx	TeamID_TeamName_DFMEA_DVP&R.xlsx
Design Improvement Document	Only for Returning Teams	.pdf	TeamID_TeamName_Design_Improvement.pdf

Note for number of Submissions: Only one submission of the Engineering Design Docket per team will be considered. Any subsequent submissions or revisions sent via email will not be accepted. Teams are therefore advised to ensure that the final version of their docket is submitted well in advance to avoid any inconvenience. Delayed submissions of the Engineering Design Docket will **not** be accepted, and such entries will **not be considered for evaluation.**

Judging Process:

- Engineering Design event will be held at the Physical Round event site for **All the participating teams of BAJA SAEINDIA 2026.**
- Students will be judged on the creation of design specifications and their ability to meet those specifications, as well as their proficiency in computer-aided drawing, analysis, testing, and development, manufacturability, serviceability, system integration, and the overall functionality of the vehicle.
- Each of these parts of the engineering product development cycle will be judged within the following subsystems: **Vehicle Overview and Integration, Rollcage & Ergonomics, Powertrain, Suspension, Steering & Brakes.**

Presentation Format:

- Teams are required to carry A1-size posters for on-site evaluation along with the 4 set of hard copies of the Design Report, Design Specification sheet, Design Improvement, DFMEA and DVP submitted for the Design Docket Evaluation.
- Each team will be evaluated over a 30 Minute period (5min for setup and introduction + 10 min of presentation + 10min of Q&A + 5min of transition and scoring). Teams must come to the evaluation area 15 minutes before their respective time slot as per the slotting released on the forum.
- Minimum No. of students allowed: 4 (1 member in each sub system).
- Maximum No. of students allowed: 12

Scoring Pattern:

Event	Marks	Phase
Engineering Design Presentation	100	Physical Round

Scoring Criteria:

The scoring criteria will be based on the team's Design Presentation, along with the evaluation of the Design Docket submitted.

Breakout	Sub-system	Evaluation Criteria	Particulars
Introduction		Team Introduction	Team ID, Team Name & Institute Name
		Vehicle Summary	Vehicle Objectives, Performance Targets, Features and Improvements, Vehicle Specifications
			Vehicle Comparison (Only for Returning Teams)
			CAD Views – Front, Side, Top & Isometric View
Vehicle Overview & Integration (Specific)		Criteria for Design	Vehicle Integration – Objectives & Methodology
			Rulebook Compliance, Lessons Learnt
		Design Calculations & Analysis	Fastner Grades & Tightening Torque
		2D & 3D Layout	Exploded View of the Vehicle
			Interface Requirements & Integration Improvements – Digital Mock-up Analysis
			Assembly Sequence of Complete Vehicle & Design for Service
		Cost & Weight Optimization	Subsystem Wise Analysis
		DFMEA/PFMEA	Vehicle Overview & Integration (Specific)

Roll Cage & Ergonomics		Criteria for Design	Objectives, Performance Targets, Features & Improvements
			Specifications, Design Comparison, Rulebook Compliance
			Lessons Learnt
		Design Calculations & Analysis	CAE Analysis
		2D & 3D Layout	CAD Views, Ergonomics, Field of Vision, Reach Envelope & Ergonomic Risk Assessment
		Cost & Weight Optimization	Rollcage System
Suspension, Steering & Brakes	**Suspension & Steering System**	Criteria for Design	Objectives, Performance Targets: Features & Improvements
			Suspension & Steering System: Specifications, Design Comparison, Design Matrix and Rulebook Compliance
			Lessons Learnt
		Design Calculations & Analysis	Suspension & Steering System
		2D & 3D Layout	
		Cost & Weight Optimization	
		DFMEA/PFMEA	
		DVP	
	Brakes	Criteria for Design	Objectives, Performance Targets: Features & Improvements, etc.
			Brakes System: Specifications; Design Comparison; Design Matrix, Rulebook Compliance
			Lessons Learnt
		Design Calculations & Analysis	Brakes System
Powertrain mBAJA		Criteria for Design	Objectives, Performance Targets; Features & Improvements
			Specifications; Design Comparison; Design Matrix; Rulebook Compliance
			Lessons Learnt
		Design Calculations & Analysis	Traction Analysis
		2D & 3D Layout	4WD/AWD – Layout & Features
			Virtual Vehicle Validation
		Cost & Weight Optimization	Powertrain System
		DFMEA/PFMEA	
		DVP	

Powertrain eBAJA	**Powertrain**	Criteria for Design	Objectives, Performance Targets; Features & Improvements
			Specifications; Design Comparison; Design Matrix, Rulebook Compliance
			Lessons Learnt
		Design Calculations & Analysis	Powertrain System
			Traction Analysis
		2D & 3D Layout	4WD/AWD/2WD – Layout & Features
			Virtual Vehicle Validation
		Cost & Weight Optimization	Powertrain System
		DFMEA/PFMEA	
		DVP	
Electricals	**Electricals**	Criteria for Design	Objectives, Performance Targets; Features & Improvements, etc.
			Specifications; Design Comparison; Design Matrix; Rulebook Compliance
			Lessons Learnt
			HV & GLV System – Starting Circuit, Components & Circuit Diagram
		Design Calculations & Analysis	Component Analysis, Circuit Analysis, Wiring Harness & its Sizing
		2D & 3D Layout	Electricals – Layout & Features
			Virtual Vehicle Validation
		Cost & Weight Optimization	Electricals System
		DFMEA/PFMEA	
		DVP	